

Advanced Data Analysis



Fit a regression predicting `exam_score` from `hours_studied` and tell me about the relationship, including anything I should be careful about.



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I fit a simple linear regression predicting `exam_score` from `hours_studied`.

The estimated model is:

$$\text{exam_score} = 6.41 + 0.61 (\text{hours_studied})$$

Interpretation

- The coefficient for `hours_studied` is about **0.61**.
- This means that, on average, each additional hour studied is associated with about a **0.61-point increase** in exam score.
- The relationship is **positive and statistically significant**:
 - $t = 4.30$
 - $p < 0.001$

So there is strong evidence that students who study more tend to score higher.

Model fit

- $R^2 \approx 0.24$

This means the model explains about **24% of the variation** in exam scores. That's meaningful, but it also suggests that a large portion of score variation is driven by other factors not included in the model (ability, prior preparation, sleep, test anxiety, etc.).

Things to be careful about

1. The relationship is not extremely strong

Even though the effect is statistically significant, the relatively modest R^2 means predictions for individual students may still be fairly noisy.

2. Residuals do not look perfectly normal

The regression diagnostics showed:

- noticeable skewness
- heavy tails / possible outliers

That matters because:

- confidence intervals and p-values can become less reliable if assumptions are violated
- a few unusual observations may influence the fitted line

With this sample size ($n = 61$), the regression is still reasonably robust, but it would be good practice to:

- inspect residual plots
- check for influential points (e.g., Cook's distance)
- consider robust standard errors if inference is important

3. Correlation is not causation

This model shows association, not proof that studying more directly causes higher scores. There may be confounding variables involved.

Bottom line

There is a clear positive relationship between hours studied and exam performance, but:

- the effect size is moderate,
- the model leaves substantial unexplained variation,
- and diagnostic checks suggest possible non-normality or influential observations that deserve attention before making strong inferential claims.



